

Theoretical Foundations of Geometry-Spectrum-Adaptive Modal Müller Compression

Executive summary and strongest theoretical reason

Plain-language answer

The strongest theoretical reason to expect the idea to work is not merely that the boundary is smooth or that Fourier methods are spectrally accurate. It is that, in the equal-permeability scalar transmission problem, the Müller formulation subtracts interior and exterior boundary operators whose leading singular/pseudodifferential parts are the same. Those cancellations make the nonidentity part of the transmission operator substantially more regular than the individual Helmholtz layer operators. Boubendir, Domínguez and Turc state this explicitly in their Theorem 2.1: on a smooth closed curve, $S_{k_e} - S_{k_i}$ gains three Sobolev orders and $N_{k_e} - N_{k_i}$ gains one, while K_k and $K_k^\top$ themselves gain three orders. Their classical second-kind transmission system, for transmission coefficient $\nu = 1$, is exactly the unweighted version of the Müller system in the question. ¹

That observation matters because smoothing periodic operators have Fourier representations with rapidly decaying coefficients. Periodic pseudodifferential calculus gives the complementary statement that, for a symbol $a(t, n)$, the modal matrix is obtained from the Fourier coefficients of a in the coordinate variable t ; smoothness in t therefore gives rapid decay with modal separation $|m - n|$. Saranen-Wendland, McLean, Turunen-Vainikko, and Ruzhansky-Turunen provide the Fourier-series framework for pseudodifferential operators on closed curves and tori. ²

There is also a much closer numerical precedent than generic “spectral BEM”: Cai and Xu proved that a dense Fourier matrix representing a compact smooth-kernel operator can be compressed to $O(n \log n)$ nonzeros while retaining optimal convergence for a class of singular integral equations; Jiang, Wang and Yu later treated Helmholtz boundary integral equations specifically, splitting the operator into a convolution part and a compact part and reporting an $O(n)$ -nonzero compressed Fourier system, with exponential convergence under additional assumptions. ³ The 2025 Fang-Jiang-Su paper pushes the same philosophy to non-axisymmetric toroidal surfaces by proving directional Fourier-coefficient decay and a stable $O(N \log^2 N)$ -nonzero truncation. ⁴

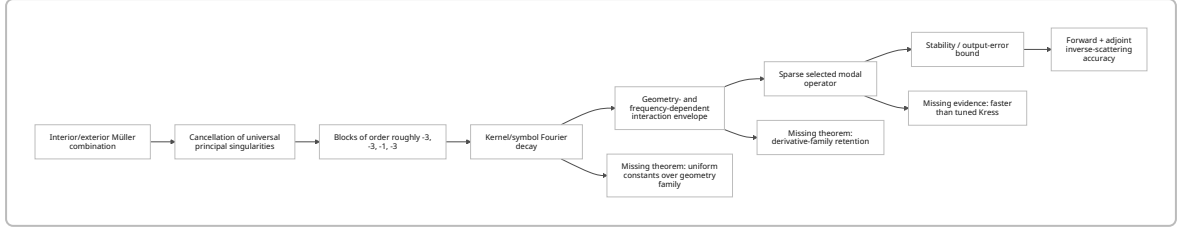
So the intellectual case is **real and strong**, but it stops short of the proposed result. Existing theory does **not** show that a geometry-spectrum-selected subset of (m, n) interactions remains uniformly accurate as a general Fourier/Laurent dielectric boundary changes during inversion. It does **not** give a selection law in terms of $K_\gamma, \{c_p\}$, electrical size and tolerance. It does **not** show that derivative matrices inherit the same compression uniformly. And it certainly does not imply an end-to-end speed advantage over a tuned Kress-Nyström solver. Those are the genuinely unresolved parts.

The most important structural conclusions are:

Conclusion	Assessment
The equal-permeability Müller differences cancel leading singular symbols and are smoothing relative to the individual layer operators.	Established
Trigonometric discretization plus analytic singularity extraction is classical and high-order/superalgebraic on smooth/analytic curves.	Established
Smooth/smoothing periodic operators have rapidly decaying Fourier representations, but the relevant decay can be in $ m + n $ or in $ m - n $, depending on whether one is exploiting a smooth bivariate kernel or a pseudodifferential symbol.	Established
Related Helmholtz BIEs can be split and Fourier-compressed with rigorous stability/convergence guarantees.	Established , for different formulations
Your $q = Ju_N$ weighting is especially favorable: it removes several artificial Jacobian/normal denominators and makes the leading coordinate symbols simpler.	Defensible extension
K_γ alone predicts compressibility.	False in general
Geometry coefficient amplitudes, derivative-weighted coefficient moments, $J_{\min}$, chord separation/reach, analytic continuation width, and kD jointly control useful constants.	Defensible extension; quantitative theorem still needed
First-order deformation of a circle produces mode sidebands selected by the deformation harmonic.	Defensible perturbative result , not a general sparsity theorem
The same retained interactions work uniformly for all shape derivatives needed by inversion.	Unverified hypothesis
Modal compression will make inversion faster than accuracy-matched Kress.	Unverified hypothesis; strongly implementation- and regime-dependent

A particularly important negative result emerges from the literature and the operator algebra: **one should not define the “singular part” as only a diagonal Fourier multiplier and then expect the remainder to be C^∞** . In Kress-type Helmholtz splittings, the logarithm is multiplied by a smooth but geometry- and frequency-dependent bivariate amplitude. For ΔV , cancellation removes the leading bare logarithm but leaves terms of the form $r^2 \log r$; this is precisely consistent with a gain of three Sobolev orders rather than infinite smoothing. For the hypersingular difference, Maue regularization leaves a weakly singular $k^2 G_k$ contribution and gives only one-order smoothing. Boubendir–Domínguez–Turc’s mapping theorem makes that distinction explicit. ⁵

A useful way to view the proposed paper is therefore:



The retrievable attachment in this session is the detailed research brief and conventions rather than separately accessible copies of the two named reports, so I treated the bibliographic leads in that brief as leads only and independently checked the primary literature. [filecite00turn0file00](#) Literature and publication status were checked through **21 September 2026**.

Scope, conventions, methodology, and literature landscape

The mathematical scope here is the two-dimensional scalar Helmholtz transmission problem on one smooth, regular, simple closed interface, with fixed homogeneous media and an unscaled normal-derivative transmission condition. This is the setting corresponding to the $\nu = 1$ case in Boubendir–Domínguez–Turc; in their electromagnetic terminology this is their “E-polarized” case. They assume nonmagnetic media and explicitly state $\nu = 1$ for that polarization, while $\nu = \epsilon_1/\epsilon_2$ for their other polarization. The exact identification with a particular engineering label such as “TMz” is convention-dependent; what matters for the present analysis is the boundary condition already fixed in the question: continuity of u and the unscaled $\partial_n u$. ⁶

This should not be promoted as a full GPR forward model. Layered ground, lossy dispersive media, finite antennas, antenna-ground coupling and three-dimensional effects would change both the physical operator and potentially its useful modal structure. The result sought here is a numerical-analysis result for a full-space scalar transmission problem, with GPR as motivation rather than as the theorem’s physical scope.

Fourier normalization. I use

$$e_n(t) = e^{int}, \quad \hat{f}_n = \frac{1}{2\pi} \int_0^{2\pi} f(t) e^{-int} dt, \quad f(t) = \sum_{n \in \mathbb{Z}} \hat{f}_n e^{int},$$

and

$$\langle f, g \rangle_{\mathbb{T}} = \frac{1}{2\pi} \int_0^{2\pi} f(t) \overline{g(t)} dt.$$

Thus $\{e_n\}_{n \in \mathbb{Z}}$ is orthonormal. This is consistent with the periodic Fourier convention used in the Nyström analysis of Boubendir–Domínguez–Turc up to harmless normalization factors in their displayed transforms. ⁶

For a pulled-back operator

$$(Bf)(t) = \int_0^{2\pi} \kappa(t, \tau) f(\tau) d\tau,$$

the Galerkin entry is

$$B_{mn} = \langle e_m, B e_n \rangle_{\mathbb{T}}.$$

If

$$\widehat{\kappa}_{r,s} = \frac{1}{(2\pi)^2} \int_0^{2\pi} \int_0^{2\pi} \kappa(t, \tau) e^{-irt} e^{-is\tau} dt d\tau,$$

then

$$B_{mn} = 2\pi \widehat{\kappa}_{m,-n}.$$

This elementary identity is important: **for a genuinely smooth bivariate kernel, matrix decay is naturally a statement about the two Fourier indices $(m, -n)$, not automatically about the distance $|m - n|$ from the diagonal.**

For

$$\ell(\theta) = \log \left(4 \sin^2 \frac{\theta}{2} \right),$$

the above normalization gives

$$\hat{\ell}_0 = 0, \quad \hat{\ell}_h = -\frac{1}{|h|}, \quad h \neq 0. \quad =$$

Boubendir–Domínguez–Turc use the same logarithmic periodic kernel in their Kress-type splitting and record the corresponding Fourier integral explicitly. ⁷

Boundary-operator convention. Set

$$G_k(x, y) = \frac{i}{4} H_0^{(1)}(k|x - y|),$$

and, with n pointing to the exterior,

$$(V_k \phi)(x) = \int_{\Gamma} G_k(x, y) \phi(y) ds_y,$$

$$(K_k \phi)(x) = \int_{\Gamma} \partial_{n_y} G_k(x, y) \phi(y) ds_y,$$

$$(K'_k \phi)(x) = \int_{\Gamma} \partial_{n_x} G_k(x, y) \phi(y) ds_y,$$

$$(T_k \phi)(x) = \text{f.p.} \int_{\Gamma} \partial_{n_x} \partial_{n_y} G_k(x, y) \phi(y) ds_y.$$

These correspond respectively to the $S, K, K^\top, N$ notation used by Boubendir–Domínguez–Turc. Some BEM literature instead defines the positive hypersingular operator $W = -T$; under that convention every appearance of T below must be sign-translated accordingly. Boubendir–Domínguez–Turc define the same four Helmholtz BIOs and treat all four in their trigonometric Nyström analysis. ⁸

Let e and i denote exterior and interior wavenumbers. With

$$\Delta V = V_e - V_i, \quad \Delta K = K_e - K_i, \quad \Delta K' = K'_e - K'_i, \quad \Delta T = T_e - T_i,$$

the $\nu = 1$ CFIESK equations displayed by Boubendir–Domínguez–Turc are

$$\begin{aligned}(I - \Delta K)u_D + \Delta V u_N &= u_D^{\text{inc}}, \\ -\Delta T u_D + (I + \Delta K')u_N &= u_N^{\text{inc}}.\end{aligned}$$

That is not merely “similar” to the requested Müller operator; after the coordinate change $q = Ju_N$, it is precisely

$$\mathcal{A}_q = \begin{pmatrix} I - \Delta K & \Delta V J^{-1} \\ -J \Delta T & I + J \Delta K' J^{-1} \end{pmatrix}.$$

Their Theorem 2.1 is therefore unusually direct evidence for this exact formulation. ¹

The classical ancestry is also clear. Kress and Roach introduced and analyzed Helmholtz transmission integral equations in 1978. ⁹ Kress's 1991 comparison treated Nyström, collocation and Galerkin trigonometric discretizations on smooth planar boundaries, using special treatment of logarithmic singularities; in that study the Nyström approach was the most efficient of the tested trigonometric methods, which is one reason an accuracy-matched Kress implementation is a serious baseline rather than a straw man. ¹⁰

Boubendir–Domínguez–Turc subsequently supplied modern Sobolev-space stability and convergence analysis for high-order Nyström discretizations of several second-kind Helmholtz transmission formulations. Their method uses global trigonometric approximation, explicit splitting into singular and smooth kernel components, and product quadratures for logarithmic singularities; they establish high-order convergence, with superalgebraic convergence for analytic boundaries. ¹¹ This makes two points critical to novelty: Kress-style singularity handling is already sophisticated, and spectral/high-order convergence is not itself new.

Trigonometric Galerkin methods are likewise established for dielectric scattering. Boriskina, Sewell, Benson and Nosich solved arbitrary-shape two-dimensional dielectric microcavity problems with uniquely solvable BIEs and angular-exponential Galerkin bases, analytically extracting the logarithmic singular part. ¹² Pinto, Aylwin and Jerez-Hanckes later established spectral Galerkin methods for quasi-periodic two-dimensional Helmholtz transmission in layered periodic media and proved well-posedness and superalgebraic convergence away from the appropriate exceptional frequencies. That is a strong spectral-transmission precedent, although its geometry and operator algebra differ substantially from a closed Müller interface. ¹³

On the inverse side, Le Louër published a three-dimensional spectral boundary-integral method for multiple dielectric obstacles, including Gauss–Newton reconstruction. ¹⁴ Hohage and Le Louër's closely related work on a spectrally accurate dielectric direct/inverse method, adjoints and iteratively regularized Gauss–Newton is verifiable as arXiv:2006.10830; in the searches performed for this report I did not locate a later version-of-record DOI, so it should currently be cited as a preprint rather than silently promoted to a journal article. ¹⁵

The closest compression literature is stronger than the initial lead list might suggest. Cai–Xu's 2008 paper explicitly compresses the Fourier representation of a compact smooth-kernel operator to $O(n \log n)$ entries and builds a fast singular-BIE solver around that compression. ¹⁶ Jiang–Wang–Yu's 2021 Helmholtz paper is closer still: according to the verified published abstract, it splits a Helmholtz integral operator into a convolution operator and a compact operator, compresses the dense Fourier matrix to $O(n)$ nonzeros, preserves stability and optimal convergence, and obtains exponential convergence under suitable assumptions. ¹⁷ Therefore a prospective paper cannot credibly claim novelty merely for “compressing a Helmholtz BIE in the Fourier basis.”

A further correction to the starting leads is useful. Fang, Jiang and Su's 2024 arXiv manuscript was titled *A Fast Fourier-Galerkin Method for Solving Boundary Integral Equations on Torus-Shaped Surfaces*. A peer-reviewed version has since appeared as *A fast Fourier-Galerkin method for solving boundary integral equations on non-axisymmetric toroidal surfaces*, **Applied Numerical Mathematics** **218** (2025), **73–90**, DOI 10.1016/j.apnum.2025.07.013. The journal paper proves a geometry-driven directional Fourier decay estimate after a shear transformation, stable truncation and quasi-optimal convergence with $O(N \log^2 N)$ nonzeros. ¹⁸

Fourier compressibility: what the theory actually says

The central conceptual mistake to avoid is treating all “rapid Fourier decay” as the same phenomenon. There are at least three different structures relevant here.

Smooth bivariate kernel. If the residual operator has

$$(Rf)(t) = \int_{\mathbb{T}} b(t, \tau) f(\tau) d\tau$$

with $b \in C^r(\mathbb{T}^2)$, repeated integration by parts gives, schematically,

$$|R_{mn}| \lesssim C_{a,b} \langle m \rangle^{-a} \langle n \rangle^{-b}, \quad a + b \leq r,$$

where $C_{a,b}$ contains norms of $\partial_t^a \partial_\tau^b b$. For C^∞ kernels the coefficient decay is superalgebraic in the **absolute pair of indices**; for a kernel analytic in a product strip of widths ρ_t, ρ_τ ,

$$|R_{mn}| \lesssim C e^{-\rho_t |m| - \rho_\tau |n|}.$$

This is elementary Fourier analysis. It naturally supports corner/hyperbolic-cross or magnitude-based retention patterns, not necessarily a narrow diagonal band.

Periodic pseudodifferential operator. In toroidal form,

$$(Af)(t) = \sum_{n \in \mathbb{Z}} a(t, n) \hat{f}_n e^{int}.$$

The Fourier matrix is exactly

$$A_{mn} = \widehat{a(\cdot, n)}_{m-n}.$$

Thus, if a is an order- μ symbol and has r controlled derivatives in t ,

$$|A_{mn}| \lesssim C_r \langle n \rangle^\mu \langle m - n \rangle^{-r}.$$

For C^∞ dependence on t , the matrix is superalgebraically localized away from the modal diagonal; for analytic t -dependence in a strip of width ρ , one obtains the schematic envelope

$$|A_{mn}| \lesssim C \langle n \rangle^\mu e^{-\rho |m-n|}.$$

The global Fourier-series representation of periodic pseudodifferential operators is classical; Saranen–Wendland gave an elementary Fourier-series treatment on closed curves, McLean established equivalence of the global Fourier and local pseudodifferential descriptions, and later periodic/toroidal calculi developed composition and symbol analysis systematically. ²

Exact convolution or symmetry. If a is independent of t , or equivalently a kernel depends only on $t - \tau$, then the Fourier matrix is exactly diagonal. This is what happens for rotationally invariant layer operators on a circle in the uniform angular parametrization. Betcke, Phillips and Spence emphasize that standard acoustic boundary operators diagonalize in the simple Fourier basis on the unit circle; on ellipses they can obtain special spectral decompositions, while on a general smooth domain they obtain only approximate transplanted-circle decompositions and a normal component plus smooth perturbation. ¹⁹ Their result is therefore valuable intuition for “diagonal plus regular perturbation,” but **it is not a sparse-matrix theorem**.

These distinctions answer several of the research questions directly:

Statement	What it actually implies
$b(t, \tau)$ is smooth	Decay of two-dimensional Fourier coefficients, typically with $ m $ and $ n $
A is a smooth periodic PDO	Off-diagonal decay in $ m - n $, weighted by operator order and symbol seminorms
Boundary is analytic	Potentially exponential coefficient decay, but only with constants controlled uniformly in a complex strip
Galerkin solution converges spectrally	Approximation error is small; the Galerkin matrix need not be sparse
Circle operator is diagonal	Exact symmetry result; it does not persist exactly under general deformation
A matrix has small entries	Does not by itself imply small operator norm after many entries are discarded
A matrix is sparse	Does not imply sparse LU factors or faster factorization
Off-diagonal blocks are low rank	Different compressibility mechanism from entrywise sparsity

This is why “spectral accuracy” and “compression” must be kept logically separate. Kress’s trigonometric methods and the modern Nyström transmission analyses can achieve very rapid convergence without asserting sparse modal matrices. ²⁰ Conversely, Cai–Xu and Jiang–Wang–Yu require an additional operator-splitting and truncation argument to move from accuracy to computational sparsity. ³

What should control the constants?

The geometry cutoff K_γ alone cannot.

A particularly simple counterexample is the ellipse

$$\gamma_\varepsilon(t) = (\cos t, \varepsilon \sin t).$$

Its Cartesian Fourier cutoff is fixed at $K_\gamma = 1$ for every $\varepsilon > 0$, yet

$$J_{\min} = \varepsilon$$

and its maximum curvature diverges like $1/\varepsilon^2$ as $\varepsilon \rightarrow 0$. The complex zeros associated with

$$J^2(t) = \sin^2 t + \varepsilon^2 \cos^2 t$$

also approach the real parameter axis as the ellipse degenerates. Thus a fixed, minimal geometry bandwidth is compatible with arbitrarily bad regularity constants and an arbitrarily small useful analytic strip. This alone rules out a meaningful compression theorem whose constants depend only on K_γ .

A finite Fourier curve does give useful **upper** derivative bounds. For example,

$$\|\gamma^{(r)}\|_\infty \leq \sum_{|p| \leq K_\gamma} |p|^r |c_p| =: G_r.$$

The weighted moments G_r are therefore more informative than the cutoff alone. But they do not supply lower bounds such as $J_{\min} > 0$, a chord-arc constant, or a minimum nonlocal separation.

A defensible geometry envelope would involve at least

$$G_r = \sum_p |p|^r |c_p|, \quad J_{\min}, \quad J_{\max}, \quad \|\kappa_\Gamma\|_\infty,$$

plus a quantitative global embedding/separation measure such as

$$c_{\text{chord}} = \inf_{t \neq \tau} \frac{|\gamma(t) - \gamma(\tau)|}{|\sin((t - \tau)/2)|} > 0.$$

The latter controls whether the standard periodic extraction

$$|\gamma(t) - \gamma(\tau)|^2 = 4 \sin^2 \frac{t - \tau}{2} D(t, \tau)$$

has a smooth, nonvanishing residual factor D . For a theorem uniform over a geometry family, the relevant quantities need **uniform** lower and upper bounds, not merely pointwise regularity at each geometry.

For analytic estimates, the more fundamental quantity is an analytic-continuation width. Even though $x(t)$ and $y(t)$ are finite trigonometric polynomials and hence entire as complex functions of t , quantities entering the kernels include

$$J(t) = \sqrt{x'(t)^2 + y'(t)^2},$$

normalized normals, logarithms of pair distances, and square roots of desingularized chord functions. Their nearest complex zeros or branch points can lie arbitrarily close to the real axis. Thus “finite Fourier geometry” does **not** mean “uniformly entire boundary operator.”

Electrical size enters independently. Differentiating a Helmholtz kernel with respect to t repeatedly generates factors involving $k\gamma'$, $k\gamma''$, and derivatives of Bessel/Hankel functions. Consequently the constants in either a C'' or analytic coefficient estimate are not generally uniform as $kD \rightarrow \infty$. The analytic strip may stay geometrically fixed while the supremum of the analytically continued oscillatory kernel grows with kD , so the effective number of significant modal couplings can still increase with frequency. Nothing in periodic pseudodifferential calculus gives a frequency-independent compression width for free. Recent high-frequency analysis of trigonometric Helmholtz BIE discretizations is encouraging about avoiding a classical pollution effect, but that is a statement about approximation versus wavelength, not about matrix bandwidth remaining bounded as k grows. ²¹

Parametrization is part of the phenomenon

The physical curve and its modal matrix are not the same object. A circle in uniform polar parameter has convolution kernels and exact Fourier diagonalization. Reparametrize the *same circle* nonuniformly and the Fourier representation generally becomes nondiagonal because the coordinate symbol acquires t -dependence. Betcke–Phillips–Spence’s exact circle diagonalization is therefore a symmetry statement for an appropriate coordinate, not a coordinate-free sparsity statement. ²²

Accordingly, diameter, physical curvature, reach and physical wavelength are geometric or physical quantities; K_γ , the c_p , $J(t)$ as a function of the chosen t , Fourier coefficient tails and modal interaction width are partly **coordinate descriptors**. A future selection rule should state explicitly whether it is invariant, approximately invariant under a restricted class of reparametrizations, or intentionally exploits a preferred coordinate. Arclength-like parametrization is theoretically attractive because it suppresses one source of unnecessary symbol variation, although the proposed Cartesian Fourier shape coordinates may make exact arclength parametrization inconvenient.

Evidence table

Claim needed for the proposed method	Closest primary result	Actual assumptions/result	Applicability here	Missing step	Status
Müller transmission operator is identity plus regularizing terms	Boubendir–Domínguez–Turc, Theorem 2.1 and Eq. (2.10)	Smooth closed curve; $S_k : H^s \rightarrow H^{s+1}$, $N_k : H^s \rightarrow H^{s-1}$, $K, K^\top : H^s \rightarrow H^{s+3}$; $S_{\kappa_1} - S_{\kappa_2} : H^s \rightarrow H^{s+3}$, $N_{\kappa_1} - N_{\kappa_2} : H^s \rightarrow H^{s+1}$	Exact unweighted match when $\nu = 1$	Transfer estimates through $q = Ju_N$ and quantify constants	Small step
Log singularity can be extracted accurately in trigonometric discretization	Kress 1991; Boubendir et al.	Smooth/analytic curves; global trigonometric interpolation; kernel splitting and product quadrature	Directly relevant	None for basic discretization; this is baseline technology	None
Smooth periodic operator has rapidly decaying Fourier representation	Saranen–Wendland; McLean; Turunen–Vainikko; Ruzhansky–Turunen	Periodic PDO symbol classes	Direct after pullback	Need block-specific symbol and uniform geometry/frequency seminorms	Significant

Claim needed for the proposed method	Closest primary result	Actual assumptions/ result	Applicability here	Missing step	Status
Dense Fourier BIE matrix can be safely truncated	Cai-Xu 2008	Their class of singular equations; compact smooth-kernel component; $O(n \log n)$ retained entries	Strong conceptual precedent	Prove analogous split/ retention for four-block Müller operator	Significant
Helmholtz BIE admits convolution-plus-compact Fourier compression	Jiang-Wang-Yu 2021	Published abstract states convolution/ compact splitting, $O(n)$ sparse matrix, stability, optimal and conditionally exponential convergence	Very close PDE precedent	Not dielectric Müller; no geometry-family/shape-derivative result	Significant
Analytic geometry can induce explicit matrix-entry decay	Fang-Jiang-Su 2025	Smooth non-axisymmetric toroidal surfaces; directional analytic continuation; $O(N \log^2 N)$ retained entries	Shows geometry-specific analytic continuation can drive selection	Different dimension/ topology/ kernel; no inverse derivatives	Significant
General smooth domain is “circle diagonal plus regular perturbation”	Betcke-Phillips-Spence 2014	Acoustic BIOs on smooth domains; approximate spectral decompositions; smooth perturbation	Good conceptual support	Does not supply sparse truncation or transmission-family bounds	Significant
K_γ alone predicts decay	None; simple ellipse counterexample refutes it	Fixed K_γ can coexist with $J_{\min} \rightarrow 0$, curvature blow-up	Directly relevant	Replace by quantitative regularity/ analyticity parameters	—

Claim needed for the proposed method	Closest primary result	Actual assumptions/ result	Applicability here	Missing step	Status
Same decay remains after shape differentiation	Potthast; Costabel–Le Louër establish differentiability, not coefficient compression	Smooth deformations; pseudo-homogeneous kernels; derivatives exist in Sobolev spaces	Supplies calculus needed for proof	Uniform Fourier estimates depending on direction harmonic	Major
Compression preserves inverse gradients	General perturbation/ adjoint principles	Requires state/ adjoint stability and controlled residual/model error	Directly adaptable	Goal-oriented estimator specific to retained interactions	Moderate
Compression is faster than tuned Kress	No theorem applicable to this comparison	Depends on selective assembly, sparse solver/ fill, RHS count, frequency and geometry updates	Must be measured	End-to-end complexity and experiments	Major

Transfer to the weighted Müller blocks

This is the point where the proposal becomes more compelling than a generic Fourier-compression argument.

Boubendir–Domínguez–Turc's Theorem 2.1, on the printed page containing their theorem, states for a smooth closed Γ and every $s \in \mathbb{R}$

$$S_k : H^s(\Gamma) \rightarrow H^{s+1}(\Gamma), \quad N_k : H^s(\Gamma) \rightarrow H^{s-1}(\Gamma),$$

$$K_k, K_k^\top : H^s(\Gamma) \rightarrow H^{s+3}(\Gamma),$$

and, for distinct wavenumbers,

$$S_{\kappa_1} - S_{\kappa_2} : H^s \rightarrow H^{s+3}, \quad N_{\kappa_1} - N_{\kappa_2} : H^s \rightarrow H^{s+1}.$$

This is the most load-bearing theorem in the literature bridge. ²³

Translated to the present notation,

$$\Delta V \text{ is order approximately } -3, \quad \Delta T \text{ is order approximately } -1,$$

while K_e, K_i, K'_e, K'_i are themselves strongly smoothing on a smooth planar curve. Thus, schematically,

$$\mathcal{A} - I \sim \begin{pmatrix} \Psi^{-3} & \Psi^{-3} \\ \Psi^{-1} & \Psi^{-3} \end{pmatrix}$$

before accounting for smooth weights and the precise coordinate pullback. That is considerably more favorable for Fourier compression than an arbitrary four-block Calderón operator containing an uncanceled order-+1 hypersingular block. ²⁴

Why the $q = Ju_N$ variable is particularly helpful

Pull back V_k to the parameter:

$$(V_k \phi)(t) = \int_0^{2\pi} G_k(\gamma(t), \gamma(\tau)) \phi(\tau) J(\tau) d\tau.$$

Then

$$(V_k J^{-1} q)(t) = \int_0^{2\pi} G_k(\gamma(t), \gamma(\tau)) q(\tau) d\tau.$$

The source Jacobian disappears exactly.

Similarly,

$$(K_k u)(t) = \int_0^{2\pi} \nabla_y G_k(\gamma(t), \gamma(\tau)) \cdot [J(\tau) n(\tau)] u(\tau) d\tau.$$

Writing

$$r(\tau) = J(\tau) n(\tau),$$

the normalized normal has disappeared in favor of the unnormalized normal vector r , which is simply a 90° rotation of γ' , up to the orientation convention.

For the adjoint block,

$$(JK'_k J^{-1} q)(t) = \int_0^{2\pi} r(t) \cdot \nabla_x G_k(\gamma(t), \gamma(\tau)) q(\tau) d\tau.$$

Again, neither a source J^{-1} nor an output $1/J$ survives. This does not prove sparsity, but it removes a potentially unpleasant independent Fourier spectrum associated with J^{-1} .

The hypersingular block is even more revealing. For the convention

$$T_k = \partial_n D_k,$$

the closed-curve Maue identity can be written, with signs adjusted to this definition,

$$T_k u = \partial_s V_k(\partial_s u) + k^2 \int_{\Gamma} G_k(\cdot, y) (n(\cdot) \cdot n(y)) u(y) ds_y.$$

Maue-type integration by parts is the standard way to replace the finite-part hypersingular expression by weakly singular kernels and tangential differentiation; modern Nyström analyses likewise treat the hypersingular BIO through such regularized structures. ²⁵

Multiplying by the target Jacobian and using $\partial_s = J^{-1}\partial_t$ gives the particularly clean coordinate formula

$$(JT_k u)(t) = \partial_t \int_0^{2\pi} G_k(\gamma(t), \gamma(\tau)) \partial_\tau u(\tau) d\tau + k^2 \int_0^{2\pi} G_k(\gamma(t), \gamma(\tau)) \gamma'(t) \cdot \gamma'(\tau) u(\tau) d\tau$$

for a consistently oriented planar curve. This formula is a derivation in the present convention; a source defining $W = -T$ has its overall hypersingular sign reversed.

This identity explains both the opportunity and the limitation. For the difference,

$$J\Delta T = \partial_t \Delta V_0 \partial_t + [k_e^2 G_{k_e} - k_i^2 G_{k_i}] (\gamma'(t) \cdot \gamma'(\tau))$$

schematically, where ΔV_0 denotes the unweighted kernel difference. Since $\Delta G = G_{k_e} - G_{k_i}$ has its leading logarithm cancelled, the first term is more regular; but the second contains

$$(k_e^2 - k_i^2) \log |t - \tau|$$

at leading weak-singular order. Hence the Müller hypersingular difference is **not an everywhere-smooth kernel after merely cancelling the most singular hypersingular symbol**. Its net order is -1 , exactly as Boubendir–Domínguez–Turc's theorem says. ¹

What singular splitting should mean here

Near the diagonal,

$$G_k(r) = -\frac{1}{2\pi} \log r + C_k + O(r^2 \log r).$$

For a periodic parametrization,

$$\log |\gamma(t) - \gamma(\tau)| = \frac{1}{2} \ell(t - \tau) + \frac{1}{2} \log D(t, \tau),$$

with

$$D(t, \tau) = \frac{|\gamma(t) - \gamma(\tau)|^2}{4 \sin^2((t - \tau)/2)}.$$

Kress-type kernel splitting does not retain only the bare $\ell(t - \tau)$: the logarithm is multiplied by smooth Bessel/geometry amplitudes, and those amplitudes can mix Fourier modes. Boubendir–Domínguez–Turc explicitly describe their discretization in terms of a singular/smooth split and logarithmic product quadrature rather than a geometry-independent diagonal singular matrix. ²⁶

This validates the caution in the research brief. A defensible decomposition is more like

$$\kappa_{\alpha\beta}(t, \tau) = a_{\alpha\beta}(t, \tau; k, \gamma) \ell(t - \tau) + b_{\alpha\beta}(t, \tau; k, \gamma),$$

possibly together with differentiated convolution terms from Maue's identity. Then

$$A_{mn}^{\text{sing}}$$

contains the Fourier convolution of the bivariate amplitude a with the known coefficients $-1/|h|$. Unless a depends only on $t - \tau$, this is not diagonal.

For the Müller *differences*, however, many amplitudes vanish to additional order on the diagonal. For example,

$$J_0(k_e r) - J_0(k_i r) = O(r^2),$$

so the log-amplitude in ΔV has a quadratic zero. This is the kernel manifestation of the three-order smoothing result. It suggests that the protected contribution itself may be highly structured and compressible, but that requires a different proof from simply declaring it “singular and dense.”

A particularly clean theoretical strategy would therefore be:

$$A_M = I + A_M^{(-1)} + A_M^{(-3)},$$

where the order—1 component is an explicitly regularized representation of $J\Delta T$, and the remaining transmission blocks are treated as order—3 operators, with any exact convolution pieces retained analytically. That decomposition aligns more directly with the known operator orders than a generic “singular plus smooth remainder” label.

A small-deformation circle calculation

Let

$$Z_0(t) = ae^{it}.$$

In the uniform angular coordinate all four circle BIOs are rotationally invariant; their kernels depend only on $t - \tau$, so the modal matrices are diagonal. This is an exact symmetry statement and is consistent with the circle results emphasized by Betcke–Phillips–Spence. ²²

Now let the physical normal displacement be

$$h(t) = h_p e^{ipt}$$

and differentiate a rotationally equivariant operator at the circle. At first order, every term in the derivative kernel is linear in $h(t)$, $h(\tau)$ or their derivatives, multiplied by a function of $t - \tau$. With the present Fourier convention,

$$\langle e_m, A'[h_p]e_n \rangle = 0 \quad \text{unless} \quad m - n = p.$$

For a real deformation $\cos pt$ or $\sin pt$, the first sidebands are therefore

$$m = n \pm p.$$

This is a **first-order perturbative selection rule**, not a global bandedness theorem. At second order combinations $p_1 + p_2$ appear; repeated nonlinear interaction spreads support further. Strong deformation can therefore destroy a narrow sideband picture even if each geometry coefficient was introduced individually.

It is also essential not to identify p above with a Cartesian Laurent coefficient index blindly. For the base circle, the outward unit normal is represented by e^{it} . A Cartesian perturbation

$$\delta Z(t) = d_q e^{iqt}$$

has physical normal component

$$h(t) = \operatorname{Re}[\delta Z(t)e^{-it}] = \operatorname{Re}[d_q e^{i(q-1)t}].$$

Thus a real parameter controlling c_q produces, at the circle, a physical normal harmonic of order $|q - 1|$, together with a tangential component. The latter affects a pulled-back parametrized matrix even though it is a reparametrization direction rather than a first-order physical shape change. This is precisely why Cartesian geometry indices, physical deformation harmonics and modal coupling indices must remain distinct.

Where the transfer can fail

Smoothness alone does not prevent the compression constants from becoming useless. The mechanism is expected to deteriorate when $J_{\min}$ becomes small, curvature becomes large, different portions of the boundary nearly touch, or the nearest complex singularity of the parametrized geometry approaches the real parameter axis. Those effects can occur without increasing K_γ .

High electrical size is another failure mechanism. Even an analytic curve can support strongly oscillatory kernels and traces. The nominal analytic decay rate in modal separation may coexist with a frequency-dependent prefactor or effective bandwidth large enough to erase practical sparsity. The recent Helmholtz Fourier-Galerkin compression results establish useful compression for their operator class but do not supply a frequency-uniform theorem for the present Müller family. ¹⁷

Finally, poor conditioning amplifies any operator truncation. Boubendir–Domínguez–Turc were motivated partly by the substantial performance differences between alternative second-kind transmission formulations, particularly in more demanding regimes; second-kind structure is valuable but does not mean the inverse norm is uniformly bounded over all frequency, contrast and geometry. ²⁷

Shape differentiation, inversion, and error control

The existing shape-calculus literature supplies the right differentiability machinery, but not the compression result that the paper would need.

Potthast's 1994 paper is a foundational boundary-integral differentiability result for inverse acoustic scattering, establishing Fréchet differentiability of boundary integral operators with respect to boundary perturbations. The verified citation is *Inverse Problems* 10 (1994), 431–447, DOI 10.1088/0266-5611/10/2/016. ²⁸ Kirsch's 1993 domain-derivative work similarly characterizes boundary perturbations through a new scattering problem and is foundational for inverse-scattering Newton methods.

Costabel and Le Louër provide the strongest modern operator-calculus precedent. Part I studies shape differentiability of pseudo-homogeneous boundary integral operators on smooth deformations and establishes infinite differentiability for broad weakly singular classes without a loss of regularity; strongly/hypersingular operators are treated through decompositions involving weakly singular operators and surface differential operators. Part II applies that framework to electromagnetic scattering by a homogeneous dielectric obstacle. ²⁹ Hettlich's 2012 penetrable-interface paper proves existence of the domain derivative and characterizes it through a transmission boundary-value problem. ³⁰

Those results justify statements of the form

$$\eta \mapsto A(\eta)$$

is differentiable, even C^∞ , after suitable pullback to a reference boundary. They do **not** justify

$$\partial_\eta A \quad \text{has the same practically small Fourier interaction set as } A.$$

The missing estimate is quantitative. If a geometry direction is a Fourier harmonic

$$h_p(t) = e^{ipt},$$

then

$$\|\partial_t^r h_p\|_\infty = |p|^r.$$

Every shape derivative of a kernel contains combinations of the deformation direction, its tangential derivatives, changes in normals and Jacobians, and differences such as $h(t) - h(\tau)$. Thus a bound on the differentiated kernel naturally contains factors growing with the harmonic order p . Differentiability therefore says that every fixed direction is legitimate; it does not imply uniform compressibility as $|p|$ grows.

This also explains why

$$|c_p| \ll 1$$

does not imply

$$\|\partial_{\text{Re } c_p} A\| \ll 1.$$

A parameter derivative removes the small coefficient and replaces it by a unit direction e^{ipt} ; derivative norms may in fact grow polynomially with $|p|$. Likewise, a small current matrix entry A_{mn} can have a large geometry derivative if the entry crosses zero or if geometry perturbation shifts significant spectral content into that interaction.

Pulled-back derivative versus physical shape derivative

There are three different objects that should not be conflated:

$$\frac{\partial}{\partial \eta} \text{ (operator pulled back to the fixed } t\text{-circle) },$$

the physical Hadamard shape derivative determined by the normal deformation

$$h = V \cdot n,$$

and the derivative of the already discretized/compressed matrix.

A physical shape functional is insensitive, at first order, to a purely tangential deformation because that deformation changes parametrization rather than the physical set. A pulled-back Fourier matrix is not generally insensitive: tangential motion changes $J(t)$, coordinate placement and how the fixed basis e^{int} corresponds to physical arclength. Costabel–Le Louër's reference-surface machinery is precisely the right conceptual framework for keeping these transport effects explicit. 31

For a fixed interaction set $\mathcal{I}$,

$$\tilde{A}(\eta) = A^{\text{sing}}(\eta) + \Pi_{\mathcal{I}} R(\eta)$$

does indeed satisfy

$$\partial_{\eta_j} \tilde{A} = \partial_{\eta_j} A^{\text{sing}} + \Pi_{\mathcal{I}} \partial_{\eta_j} R.$$

But when

$$\mathcal{I} = \mathcal{I}(\eta)$$

is selected by a hard threshold, the map is generally nonsmooth when entries enter or leave the set. “Differentiate the compressed discrete model” then differs from “compress the derivative of the continuous model.” A defensible inverse algorithm must either keep the interaction set locally fixed, employ hysteresis/smooth selection, or explicitly analyze switching errors.

Operator error to state and receiver error

Let

$$Ax = b, \quad \tilde{A} = A + E.$$

If

$$\beta = \|A^{-1}\|, \quad \beta\|E\| < 1,$$

the standard Banach perturbation argument gives

$$\|\tilde{A}^{-1}\| \leq \frac{\beta}{1 - \beta\|E\|},$$

and

$$\|\tilde{x} - x\| \leq \frac{\beta\|E\|}{1 - \beta\|E\|} \|x\|.$$

This is the right first stability factor: matrix coefficient decay matters only insofar as the discarded tail is small in an operator norm relative to the inverse stability constant.

For a receiver map C ,

$$y = Cx,$$

one has

$$\|\tilde{y} - y\| \leq \|C\| \|\tilde{x} - x\| + \|\tilde{C} - C\| \|\tilde{x}\|.$$

When receivers remain a positive distance from the boundary, layer-potential evaluation is smoother than the boundary trace map; this makes receiver-oriented error control potentially less demanding than full matrix-norm accuracy.

That observation suggests a stronger algorithmic target than “approximate every matrix entry to tolerance.” Goal-oriented error estimation uses the adjoint to weight a residual according to its effect on a quantity of interest. Dual-weighted-residual theory formalizes precisely this distinction between state error and output error, including for shape/free-boundary problems. ³² A geometry-adaptive Fourier

compressor could therefore select interactions to control receiver data or the inverse objective rather than enforcing a uniform Frobenius error that is much stronger than necessary.

The adjoint changes the derivative-cost argument

Suppose

$$\Phi(\eta) = \frac{1}{2} \|C(\eta)x(\eta) - d\|_W^2, \quad A(\eta)x(\eta) = b(\eta).$$

With an adjoint satisfying, schematically,

$$A^* \lambda = C^* W (Cx - d),$$

the derivative can be written

$$\partial_{\eta_j} \Phi = \text{Re} [(Cx - d)^* W C_j x + \lambda^* (b_j - A_j x)],$$

with the obvious changes when the incident or observation operator is geometry independent.

Shape-calculus formulations can go one step further and represent the derivative as a Hadamard boundary integral involving forward and adjoint traces rather than materializing an A_j matrix for every parameter. Adjoint methods are specifically valuable because they replace a number of sensitivity solves proportional to the parameter dimension by a small number of adjoint solves; BIE-based shape optimization literature uses exactly this principle. ³³

Consequently, **derivative-matrix compression is not necessary to beat a sensible baseline**. A tuned Kress code can factor its forward Müller matrix once for each geometry/frequency and reuse that factorization for all incident fields and adjoints. It can obtain a shape gradient from reciprocal/Hadamard trace products without building one dense $\partial_{\eta_j} A$ per geometry parameter. Any cost comparison that attributes $O(PN^2)$ derivative assembly to that baseline, where P is the number of shape parameters, would be unfair.

A better theoretical target for inverse scattering is therefore:

compress enough to preserve $u_D, q, \lambda_D, \lambda_q,$ and their required boundary products,
rather than

compress every A_j uniformly.

For example, if a Hadamard gradient takes the abstract form

$$g[h] = \int_{\Gamma} h(s) G(u, \lambda)(s) ds,$$

then for a normalized deformation basis h_j ,

$$|\tilde{g}[h_j] - g[h_j]| \leq \|h_j\|_X \|\tilde{G} - G\|_{X^*}.$$

This moves the burden toward forward/adjoint trace accuracy and away from a separate matrix approximation theorem for every geometry derivative.

There is mature optimization theory allowing state and adjoint equations to be solved inexactly provided the errors are coordinated with the optimization accuracy. Ziems and Ulbrich, for example, give adaptive inexact PDE-constrained SQP frameworks with explicit accuracy requirements for state and adjoint solves; more recent discrete-adjoint analyses directly bound gradient error in terms of state/adjoint residual tolerances. ³⁴ These do not prove anything about Müller compression, but they support the methodological principle that an inverse solver need not drive the forward operator to unnecessarily stringent full-matrix accuracy at every iteration.

Advantages, limitations, and break-even conditions

The theoretical consequences need to be separated from the hoped-for computational advantages.

What follows already from theory. The $\nu = 1$ Müller nonidentity blocks are regularizing; trigonometric discretizations are high-order; Fourier representations of sufficiently smooth/smoothing operators decay; and closely related singular/Helmholtz Fourier-Galerkin matrices have been rigorously compressed while retaining stability and convergence. ³⁵

What is plausible but requires experiments. For moderate electrical size and uniformly analytic, moderately deformed boundaries, a selected set of modal interactions may be much smaller than M^2 , and that set may vary predictably with the geometry spectrum and tolerance. Repeated solves during multifrequency inversion may then benefit from reduced memory and application cost.

What remains speculative. A smaller retained set does not yet imply a faster factorization, faster geometry rebuild, or faster inverse reconstruction than Kress.

A useful accounting is:

Method/ operation per geometry and frequency	Storage	Assembly	Factorization / solve	Key caveat
Dense tuned Kress–Nyström, N nodes	$O(N^2)$	$O(N^2)$	dense LU $O(N^3)$, each reused RHS $O(N^2)$	Very mature singular quadrature; factorization reusable
Dense Fourier– Galerkin, M modes	$O(M^2)$	typically $O(M^2)$ or worse if naively quadrature- assembled	$O(M^3)$ LU	Spectral basis alone gives no complexity reduction
Full tensor kernel sampling + FFT + truncation	$O(M^2)$ during build	still essentially $O(M^2)$ samples/ work	depends on retained solver	Does not solve assembly bottleneck

Method/ operation per geometry and frequency	Storage	Assembly	Factorization / solve	Key caveat
Sparse modal matrix with S directly assembled retained entries	$O(S)$	potentially $O(S \text{ polylog} M)$	iterative $O(IS)$ per RHS	Needs stable iteration and selective quadrature
True banded matrix, half- bandwidth w	$O(Mw)$	$O(Mw)$ if direct	band LU $O(Mw^2)$, solves $O(Mw)$	Requires actual band structure, not merely decay
Arbitrary sparse matrix with S entries	$O(S)$	$O(S)$ if selective	sparse LU depends on fill	$S \ll M^2$ does not imply cheap factors
Convolution/ diagonal protected part + sparse remainder	$O(M + S)$	potentially near- linear	FFT/Krylov potentially near- linear	Closest to Cai- Xu/Jiang precedents
Kress/FMM or hierarchical- matrix upgrade	near-linear or compressed	method dependent	fast iterative/ direct alternatives	Raises the competitive bar beyond dense Kress

Let there be F frequencies, L incident fields and L_a adjoint right-hand sides per geometry. A dense direct baseline is roughly

$$C_{\text{dense}} \sim F (C_A N^2 + C_{\text{LU}} N^3 + (L + L_a) C_s N^2).$$

A sparse iterative modal realization would look more like

$$C_{\text{modal}} \sim F (C_{\text{select}} + C_{\text{retained}} S + (L + L_a) I C_{\text{mv}} S),$$

where $S = |\mathcal{I}|$ is the retained interaction count and I is the iteration count.

That comparison exposes the actual break-even questions. Is S substantially smaller than N^2 ? Can those S coefficients be constructed without first computing N^2 or M^2 data? Is the protected singular contribution itself diagonal/convolutional/sparse enough to apply cheaply? Does the resulting system have a robust preconditioner? Does I stay bounded over frequency and geometry? How often must the interaction pattern be recomputed? None of these answers follows from spectral convergence.

The same caution applies to system size. A Fourier–Galerkin formulation is not automatically lower-dimensional than a Kress method. Kress's high-order Nyström method already uses global trigonometric interpolation on a periodic boundary. Boubendir–Domínguez–Turc likewise use trigonometric approximations. ³⁶ Therefore

$$M < N$$

at a fixed accuracy is an empirical possibility, not a structural theorem.

Similarly,

“many zeros” $\nRightarrow$ “fast dense LU.”

Passing a mostly zero matrix to a dense factorization still incurs dense costs; passing it to a generic sparse direct solver can create fill. Cai–Xu’s fast complexity result is meaningful precisely because their algorithm exploits the particular compressed structure rather than merely inserting zeros. ¹⁶

Where a credible advantage exists

The most favorable regime is likely:

uniformly regular/analytic boundary family
+ low-to-moderate geometry spectrum
+ moderate kD
+ many incident/adjoint RHS
+ many geometry iterations
+ direct selective coefficient construction
+ structured sparse or iterative solver.

In that regime, the exterior–interior cancellation may leave a very compressible perturbation of the identity, geometry changes may alter the interaction envelope only gradually, and amortizing setup over many forward/adjoint solves could matter.

The least favorable regime is approximately the opposite: high kD , high-order geometric corrugation, near self-contact, strong parameter-induced reparametrization, poor conditioning, frequent major geometry changes, or small M where dense BLAS/LAPACK is already cheap. A Kress solver is also likely preferable if compression requires an M^2 oversampled kernel grid and two-dimensional FFT before every geometry update: that would preserve much of the expensive work while adding selection overhead.

The 2021 Jiang–Wang–Yu result makes the novelty bar particularly useful here. Since a Helmholtz convolution-plus-compact Fourier compression already exists, the proposed method is most compelling only if the **transmission cancellation and explicit geometry dependence** lead to a stronger or more useful structure than that generic precedent. ¹⁷

The 2025 Fang–Jiang–Su result raises a second bar: geometry-dependent analytic continuation has now been used explicitly to derive a matrix truncation rule on non-axisymmetric surfaces. ⁴ A new paper should therefore not market “using geometry to predict Fourier decay” as wholly new. Its differentiator would be a **quantitative geometry-family result for closed dielectric Müller transmission, including shape variation and preferably inverse quantities of interest.**

Paper-level contribution, prioritized recommendations, and citation backbone

The narrowest defensible new theorem

The strongest plausible paper contribution is not “Fourier–Galerkin Müller scattering” and not “spectral dielectric inversion.” Both neighborhoods are populated. It is something closer to:

A geometry- and frequency-explicit Fourier interaction-decay theorem for the weighted $\nu = 1$ Müller transmission operator, uniform over a quantitatively nondegenerate family of Fourier-parametrized boundaries, together with a stable adaptive interaction truncation.

A realistic theorem statement would assume a family $\mathcal{G}$ of embeddings satisfying, for fixed constants,

$$0 < J_* \leq J(t) \leq J^*,$$

$$c_{\text{chord}} \geq c_* > 0,$$

$$\|\gamma\|_{C^r} \leq G_r,$$

or, in the analytic version, extension to

$$|\operatorname{Im} t| < \rho$$

with uniform nonvanishing of the relevant complexified speed and chord functions. One would also restrict

$$k_e D, k_i D \leq \kappa_{\max}$$

for a frequency-uniform statement.

After an exact Kress/Maue splitting, one would aim to prove blockwise bounds such as

$$|(R_{\alpha\beta})_{mn}| \leq C_{\alpha\beta,r} \langle n \rangle^{-\sigma_{\alpha\beta}} \langle m - n \rangle^{-r},$$

with

$$\sigma_{\alpha\beta} \approx \begin{cases} 3, & DD, DN, NN, \\ 1, & ND, \end{cases}$$

in the schematic pseudodifferential sense, or an analytic counterpart

$$|(R_{\alpha\beta})_{mn}| \leq C_{\alpha\beta} \langle n \rangle^{-\sigma_{\alpha\beta}} e^{-\rho_{\text{eff}} |m-n|}.$$

The exact allocation of powers must come from the chosen Sobolev normalization and protected term; the displayed formula is a theorem **target**, not an established result.

The unresolved proof step is quantitative: turn the standard smooth-boundary mapping properties plus periodic PDO calculus into constants expressed in terms of the chosen geometry-family bounds and kD , while tracking the J -weighted pullback exactly. Neither Boubendir–Domínguez–Turc nor the general periodic PDO references supply that geometry-explicit constant bookkeeping. 37

A second, stronger target: derivative-family compression

The higher-impact result would add a shape derivative. For a physical deformation direction h ,

$$|(\partial_h R_{\alpha\beta})_{mn}| \leq C_{\alpha\beta,r} \|h\|_{C^{r_h}} \langle n \rangle^{-\sigma'_{\alpha\beta}} \langle m - n \rangle^{-r}.$$

For a harmonic direction h_p ,

$$\|h_p\|_{C^{r_h}} \sim \langle p \rangle^{r_h},$$

so the theorem would explicitly expose the price of high geometry harmonics rather than hiding it.

Such a theorem would genuinely connect Potthast/Costabel–Le Louër shape differentiability to Cai–Xu/Jiang-style Fourier compression; I found no cited precedent that makes that connection for a dielectric transmission operator. The existing shape-calculus results establish differentiability, not uniform sparse Fourier representation of the derivative family. ³⁸

A third target: certified interaction selection

Instead of prescribing a band width heuristically from K_γ , define an envelope E_{mn} satisfying

$$|(A - A_{\mathcal{I}})_{mn}| \leq E_{mn} \quad ((m, n) \notin \mathcal{I})$$

and choose $\mathcal{I}$ so that weighted row and column tail sums obey

$$\sup_m \sum_n E_{mn} \frac{w_m}{w_n} \leq \varepsilon, \quad \sup_n \sum_m E_{mn} \frac{w_m}{w_n} \leq \varepsilon.$$

A weighted Schur estimate then yields an operator-norm bound in the chosen Fourier/Sobolev scaling. Combining this with

$$\|A^{-1}\|_{\varepsilon} < 1$$

immediately provides a state-error guarantee.

For inverse scattering, an even stronger numerical contribution would replace this full-state criterion by a receiver- or adjoint-weighted estimator. The established dual-weighted-residual literature provides the conceptual machinery for targeting functionals rather than global solution norms. ³⁹

Priority order

The recommended intellectual priorities are:

1. **Prove the blockwise Fourier-decay theorem for the weighted Müller differences with quantitative geometry/frequency constants.** This is the foundational result; without it “geometry-spectrum adaptive” remains descriptive rather than mathematical.
2. **Extend it to a uniformly controlled shape family or to first derivatives.** A full derivative-family result would distinguish the work sharply from existing Helmholtz Fourier compression.
3. **Only then make a speed claim**, and make it against an accuracy-matched Kress baseline that reuses factorizations and uses adjoint/Hadamard gradients. The algorithm must compute retained entries without first constructing the dense matrix.

A weaker but still publishable path would replace the full derivative theorem with: a rigorous forward decay bound, an operator-norm adaptive truncation criterion, a first-order circle-deformation selection theorem, and numerical evidence that the same selections remain robust during shape inversion. That would need to be labeled honestly as a theorem plus experimentally validated extension, rather than as a proved uniform inverse-scattering result.

Simple falsification tests

A small number of tests can falsify the central hypothesis without becoming an implementation roadmap.

Fixed K_γ , deteriorating regularity. Use an ellipse family with aspect ratio tending to zero. If retained interaction count is controlled mainly by K_γ , it should remain stable; the theory above predicts instead that it worsens as $J_{\min}$, curvature and analytic-strip constants deteriorate.

Fixed physical shape, changed parametrization. Reparametrize a circle increasingly nonuniformly. If compression is physical-geometry driven, it should remain unchanged; Fourier PDO theory predicts potentially substantial coordinate-induced coupling.

Frequency sweep at fixed analytic geometry. Track interaction count versus kD . A frequency-independent count would be surprising; systematic growth would quantify the missing k -dependence.

Derivative stress test. Compare the forward matrix tail to $\partial_{\eta_p} A$ for increasing $|p|$. Uniform retention despite increasing p is the optimistic hypothesis; shape calculus suggests constants should worsen with harmonic order.

Assembly accounting. Compare “selected direct assembly” against “full grid then truncate.” If almost all speedup disappears in the latter, the method’s true contribution is a selective assembly rule rather than matrix compression per se.

Manuscript-style motivation and related-work passage

A supported version could read:

Boundary-integral formulations of smooth two-dimensional Helmholtz transmission problems admit high-order trigonometric discretizations, and classical Müller/Kress–Roach second-kind formulations benefit from cancellation between the interior and exterior Calderón operators. In particular, for smooth closed curves the difference of Helmholtz single-layer operators is three-order regularizing and the difference of hypersingular operators is one-order regularizing, while the double-layer and adjoint double-layer operators are themselves strongly smoothing in two dimensions. ²⁴ High-order Nyström schemes exploit periodic logarithmic kernel splittings to obtain rapidly convergent transmission solvers, so spectral convergence and singularity extraction are already well established. ⁴⁰ Separately, fast Fourier–Galerkin methods have shown that smooth compact remainders of singular integral operators can admit sparse Fourier representations with stability and approximation guarantees, including specific Helmholtz boundary-integral formulations. ³ What is not supplied by these results is a geometry-explicit truncation theory for the four coupled blocks of the weighted Müller transmission operator, uniform over deforming interfaces and suitable for shape inversion. The present direction is therefore to quantify how transmission cancellation, parametrized

boundary regularity, electrical size and geometry harmonics jointly control modal interaction decay, and to use those estimates to select only the interactions required for a prescribed forward or adjoint error.

That passage makes no priority claim about GPR and does not imply a speed advantage before one has been demonstrated.

Annotated citation backbone

Transmission BIE and high-order trigonometric discretization.

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Goal-oriented and inexact inverse computation.

K. G. van der Zee, E. H. van Brummelen and R. de Borst develop duality-based goal-oriented error estimates for shape/free-boundary problems, including both reference-domain and shape-linearization approaches. These are not scattering papers, but they are appropriate precedents for replacing full-state error control by receiver/objective control. 52

J. C. Ziemis and S. Ulbrich, *Adaptive Multilevel Inexact SQP Methods for PDE-Constrained Optimization*, **SIAM Journal on Optimization**, DOI [10.1137/080743160](https://doi.org/10.1137/080743160), gives a rigorous framework for adapting state/adjoint solution accuracy during optimization rather than requiring uniform maximum accuracy at every iteration. 53

Essential reading sequence

For a paper author, the shortest high-value sequence is:

First: Boubendir–Domínguez–Turc, especially Theorem 2.1, Eq. (2.10), their singular kernel decompositions, and the Nyström analysis. This establishes exactly what transmission cancellation gives you and prevents overclaiming the singular structure. 54

Second: Cai–Xu 2008 and Jiang–Wang–Yu 2021. These establish the novelty boundary for Fourier matrix compression and show what a real complexity theorem, rather than a decay observation, must contain. 3

Third: Saranen–Wendland plus McLean or Ruzhansky–Turunen. These give the cleanest mathematical language for deriving $|m - n|$ decay from coordinate-symbol regularity. 55

Fourth: Kress 1991 and the 2016 Nyström transmission paper as the numerical baseline literature. 56

Fifth: Potthast and Costabel–Le Louër before making any statement about inverse sensitivities or derivative compression. ³⁸

Sixth: Fang–Jiang–Su 2025 before claiming novelty for geometry-dependent Fourier interaction selection. ⁴

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Bottom-line assessment

The proposed direction is mathematically justified enough to investigate. The reason is unusually concrete: **the exact equal-permeability Müller formulation already converts the hardest individual boundary operators into regularizing exterior-interior differences, and established Fourier/PDO theory says regularizing periodic operators are precisely the objects whose Fourier matrices should decay.** Closely related fast Fourier–Galerkin methods prove that such decay can be turned into stable sparse algorithms. ³⁵

But the defensible claim today is not “the geometry spectrum determines a sparse Müller matrix.” The geometry cutoff is insufficient; the protected singular contribution is not generally diagonal; the hypersingular difference remains only one-order smoothing; frequency can widen the effective spectrum; parametrization matters; and shape derivatives introduce direction-dependent constants that can grow rapidly with geometry harmonic order.

The strongest paper would therefore make **uniformity** the novelty: uniformity over a quantitatively specified geometry family, explicit kD dependence, and—ideally—uniform control of the forward/adjoint or first-derivative family. A convincing computational paper would additionally show that the selected entries are generated without first assembling the dense operator and that the resulting structured solver beats an accuracy-matched Kress implementation after all assembly, factorization, multiple-right-hand-side, adjoint and geometry-rebuild costs are counted.

The remaining questions that materially determine the final theorem statement are: what quantitative admissible geometry class should be guaranteed during inversion; whether the target is an algebraic C^r result or the stronger uniform-analytic result; and whether the numerical contribution is intended to use sparse direct, preconditioned iterative, or convolution-plus-sparse algebra. Those choices affect the meaningful definition of “compressible” more than the nominal Fourier cutoff K_γ does.

¹ ⁵ ⁶ ⁷ ⁸ ²³ ²⁴ ²⁷ ³⁵ ³⁷ ⁵⁴ <https://arxiv.org/pdf/1404.1331>

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